

# Lab worksheet — zero values and conversions

Name: \_\_\_ · GitHub: \_\_\_

Fill this **before** you leave. Then tick the [lab sheet](#).

## A. Zero values

Write the zero value. Do not guess from another language.

| Declaration                | Type | Zero value (write it) |
|----------------------------|------|-----------------------|
| <code>var n int</code>     |      |                       |
| <code>var f float64</code> |      |                       |
| <code>var s string</code>  |      |                       |
| <code>var ok bool</code>   |      |                       |
| <code>var p *int</code>    |      |                       |

What printed when you ran `zeros.go`?

(paste)

## B. Which lines compile?

Mark **OK** or **FAIL**. If FAIL, write the fix.

| Code                                                                | OK / FAIL | Fix (if FAIL) |
|---------------------------------------------------------------------|-----------|---------------|
| <code>var a int = 3; var b float64 = 1.5; fmt.Println(a + b)</code> |           |               |
| <code>year := 2026; fmt.Println(float64(year) + 0.5)</code>         |           |               |
| <code>fmt.Println(int(3.9))</code>                                  |           |               |
| <code>var x int = 5; var y int64 = x</code>                         |           |               |
| <code>name := "Ada"; name := "Grace"</code>                         |           |               |
| <code>count := 1; count = 2</code>                                  |           |               |

## C. Operators (quick)

| Expression                         | Result | Why |
|------------------------------------|--------|-----|
| <code>10 / 3</code>                |        |     |
| <code>10 % 3</code>                |        |     |
| <code>true &amp;&amp; false</code> |        |     |
| <code>"Ada" == "ada"</code>        |        |     |

## D. Bytes vs runes

| String               | <code>len(...)</code> | <code>utf8.RuneCountInString(...)</code> |
|----------------------|-----------------------|------------------------------------------|
| <code>"café"</code>  |                       |                                          |
| <code>"café"</code>  |                       |                                          |
| <code>"සිංහල"</code> |                       |                                          |

One sentence: why is `len("café")` not 4?

## E. CLI + test (tick when done)

- `[] go run greet.go -first Ada -last Lovelace` prints a full name
- `[] cd names && go test` is green
- `[]` I can explain what `go mod init` did in one sentence (even if the full story is week 4)